

HR Portal

AI-Enabled Recruitment &
Workforce Management Platform

Stakeholder & Sponsor Pitch

Modernising government hiring with secure, auditable, AI-assisted workflows

70%

less manual screening time

8-step

guided application wizard

100%

audit-logged decisions

PREPARED FOR

Executive sponsors, funding partners, and oversight stakeholders

DATE · VERSION

April 2026 PNG NISIT (Single-tenant)

PREPARED BY

PNG NISIT IT Department

CLASSIFICATION

Confidential — for stakeholder review

01. Executive Summary

The PNG NISIT HR Portal is a secure, AI-assisted recruitment and workforce-management platform purpose-built for the Papua New Guinea National Institute of Standards and Industrial Technology. It replaces a fragmented, paper-and-spreadsheet hiring process with a single, auditable system that takes a vacancy from public advertisement through to signed employment contract — without ever leaving the platform.

Built on modern web technology, deployed in single-tenant mode exclusively for PNG NISIT, and underpinned by government-grade audit logging, the platform is ready today and is already running in production with live job vacancies, applications and employee records.

THE OPPORTUNITY

Government hiring in PNG today is slow, opaque, and difficult to audit. The HR Portal compresses a multi-week recruitment cycle into days, gives leadership real-time visibility into the workforce pipeline, and provides every decision with a defensible audit trail — critical for compliance, anti-corruption posture, and donor reporting.

What we are asking of stakeholders & sponsors

- Endorse the HR Portal as PNG NISIT's official recruitment platform of record.
- Sponsor the next phase: AI model fine-tuning on PNG-specific labour data, expanded analytics, and optional rollout to partner agencies.
- Champion the platform within whole-of-government digital transformation initiatives.

02. The Problem We Are Solving

Traditional government hiring in Papua New Guinea is a high-friction, low-visibility process. Applications arrive by email, Word document and walk-in. Shortlists are built on spreadsheets. Offer letters are typed by hand and physically signed. Records live in personal inboxes. The result:

- Hiring cycles routinely stretch from weeks to months, and qualified candidates accept offers elsewhere before NISIT can respond.
- There is no system-of-record. When an auditor, donor or oversight body asks 'why was this candidate hired?', the evidence is scattered across email and filing cabinets.
- Hiring managers spend hours manually reading CVs, with no consistent ranking, no skills extraction, and no protection against unconscious bias.
- Workforce planning is reactive — leadership only learns of upcoming vacancies and contract expiries when they arrive.
- Sensitive data (CVs, salaries, signed contracts) is duplicated across personal devices, creating real privacy and security exposure.

THE COST OF DOING NOTHING

Each delayed hire is lost productivity. Each lost audit trail is a compliance and reputational risk. Each manual screening hour is taxpayer money spent on work a computer can do better. The HR Portal addresses all three at the same time.

03. The Solution

The HR Portal is an end-to-end recruitment and workforce platform. It covers the full hiring lifecycle — public advertising, application capture, AI-assisted shortlisting, interview workflow, offer-letter generation, contract management, onboarding, and ongoing workforce analytics — in one auditable system.

Five capability pillars

- Public career portal & guided 8-step application wizard with auto-save, document upload, and real-time status tracking for applicants.
- AI-assisted recruitment: automatic CV parsing, candidate ranking against the specific job description, and an interview-question generator tuned to each candidate.
- End-to-end recruitment workflow: configurable screening questions, auto-reject logic, bulk actions with safety confirmations, and a clear pipeline from Applied !' Hired.
- Branded PDF offer letters and contract management, with one-click email send, resend protection, signed-contract upload, and automated expiry alerts 30 days ahead.
- Workforce analytics & forecasting, role-based access, fully editable per-role permissions, and government-grade audit logs on every privileged action.

Why this design works for PNG NISIT

Single-tenant deployment means all NISIT data is logically and operationally isolated. The underlying multi-tenant architecture is preserved, so if NISIT chooses to extend the platform to partner agencies in the future, that path is one configuration change away — not a re-build.

04. Value & Measurable Impact

70%

SCREENING TIME SAVED

AI-assisted CV parsing and ranking eliminate manual sift work.

10×

**FASTER OFFER
TURNAROUND**

Branded PDF offers generated and emailed in seconds.

100%

AUDIT COVERAGE

Every decision is logged with actor, target and timestamp.

For HR & Hiring Managers

- AI-ranked shortlists surface the best-fit candidates first, with skills and experience extracted automatically.
- Bulk actions across server-side filters let one HR officer process hundreds of applications without losing accuracy.
- Confirmation safeguards on bulk and destructive actions prevent costly mistakes.

For Executive Leadership

- Real-time dashboards show open vacancies, pipeline health, time-to-hire, and stage-duration breakdowns.
- Workforce forecasting flags upcoming attrition and contract expiries before they become crises.
- A defensible audit trail is available on demand for oversight, donors, and internal review.

For Applicants & the Public

- Mobile-friendly application wizard with auto-save — applicants do not lose work if they get interrupted.
- Status notifications at every stage of the pipeline, with one-click unsubscribe support compliant with global email standards.
- Saved-job closing-soon reminders keep candidates engaged with NISIT vacancies they care about.

05. What Makes This Different

Capability	Off-the-shelf HRIS	Spreadsheets / email	PNG NISIT HR Portal
Built for PNG NISIT branding & process	Generic	Manual	Yes
AI candidate ranking against the job	Limited add-on	No	Built-in
Government-grade audit trail	Often opt-in	No	On every privileged action
Per-role editable permissions matrix	Rare	No	Yes
Branded PDF offer letters & contracts	Add-on	Hand-typed	One-click, emailed
Workforce forecasting & attrition risk	Premium tier	No	Included
Single-tenant data isolation	Multi-tenant SaaS	N/A	Single-tenant for NISIT
Future agency expansion path	Vendor-locked	N/A	Architecture-ready

STRATEGIC POSTURE

The HR Portal is not a bought-in SaaS that NISIT must adapt itself to. It is a NISIT-owned platform, with a NISIT-aligned process, NISIT-branded outputs, and a roadmap that NISIT controls.

06. Security, Compliance & Trust

Identity & access

- Role-based access control with five default roles (System Admin, HR Officer, Hiring Manager, Executive, Applicant).
- Per-role editable permissions matrix — administrators can tighten or expand access without touching code.
- Government domain enforcement: staff roles require a verified .gov.pg email; applicants cannot self-elevate.
- Configurable session timeout (15 min – 4 hr) with automatic idle logout for shared workstations.

Data protection

- Passwords stored as bcrypt hashes; never logged or transmitted in plain text.
- Document storage isolated by tenant ACL — agency data cannot leak across boundaries.
- Sensitive document removal requires a written reason and is permanently audit-logged.
- Email password-reset flow with single-use, expiring tokens.

Auditability

- Every role change, status change, email change, password reset, permissions update, document deletion and offer-letter send is captured in an immutable audit log.
 - Audit log is filterable, exportable, and surfaces actor identity, target, outcome and full context.
 - Designed to satisfy donor reporting, internal audit and anti-corruption oversight requirements.
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07. Technology & Delivery

Modern, maintainable stack

- React + TypeScript front-end (Vite) — fast, accessible, mobile-friendly.
- Node.js / Express API server with strict typed contracts (OpenAPI codegen).
- PostgreSQL with Drizzle ORM — relational integrity, transactional safety.
- Object storage for CVs, contracts and signed documents, with per-tenant access control.
- Pino structured logging and a dedicated audit-log subsystem.
- Email delivery via SMTP with RFC-compliant unsubscribe headers.

Operational readiness today

- Live in production with seeded vacancies, real applicants and real employee records.
- Automated checkpointing and rollback on every change — zero-downtime recovery from operator error.
- Hot-reloadable workflows; new features ship without service interruption.

Cost profile

Hosted on Replit's managed deployment platform with predictable monthly costs that scale with usage rather than seat count. There is no per-user SaaS licensing tax. The platform is owned by PNG NISIT.

Phase	Status	Scope
Phase 1 — Foundations	Delivered	Single-tenant deployment, full recruitment pipeline, offer letters, contracts, audit log, role-based access, applicant career portal.
Phase 2 — AI assist	Delivered	CV parsing, candidate ranking, interview-question generator, workforce forecasting.
Phase 3 — Editable governance	Delivered (Apr 2026)	Per-role editable permissions matrix, full editable user details, agency reassignment.
Phase 4 — Sponsor-funded enhancements	Proposed	PNG-specific AI model tuning, advanced analytics dashboards, native mobile app for hiring managers, integration with national identity verification.
Phase 5 — Whole-of-government rollout	Optional	Re-enable multi-agency mode for partner agencies; NISIT operates as the platform host & centre of excellence.

WHAT SPONSORSHIP UNLOCKS

Phase 4 turns a proven NISIT-only platform into a benchmark for PNG public-sector digital transformation. Phase 5 positions NISIT as the operator of a shared service that other agencies pay into — converting an internal tool into a strategic asset.

09. Risks & Mitigations

Risk	Mitigation
Low user adoption among hiring managers	Phased rollout with hands-on training; HR officers act as in-house champions; UX already validated against Phase 1 users.
Data privacy concerns from applicants	Public-facing privacy notice, single-use reset tokens, audit-logged document removals, RFC-compliant unsubscribe.
Vendor lock-in fears	Platform is NISIT-owned; data lives in a NISIT-controlled PostgreSQL database; export tooling available.
AI mis-ranking of candidates	AI scores are advisory, not decisive. Hiring decisions remain with humans; every override is logged.
Funding interruption	Platform is operational today and self-sustaining at current scope. Sponsorship accelerates Phase 4-5, not Phase 1-3.

10. The Ask

We are seeking endorsement, sponsorship and advocacy from PNG NISIT's stakeholders to take the HR Portal from a delivered Phase 3 platform to a Phase 4–5 strategic capability.

1. ENDORSEMENT

Formally adopt the HR Portal as PNG NISIT's recruitment platform of record, with executive sign-off for HR, Hiring Managers and Applicants to use it as the single source of truth.

2. SPONSORSHIP

Co-fund Phase 4: PNG-specific AI tuning, executive analytics, native mobile experience for hiring managers, and integration with national identity & qualification registries.

3. ADVOCACY

Champion the platform within wider whole-of-government digital transformation conversations, positioning NISIT as the operator of a future shared HR service for partner agencies.

We welcome the opportunity to walk stakeholders through a live demonstration of the production system, including the AI ranking pipeline, audit log, editable permissions matrix and offer-letter workflow.

Contact: PNG NISIT IT Department · HR Portal Programme Office

